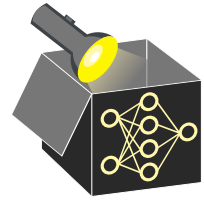
Interpretable Machine Learning

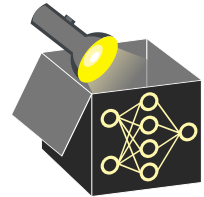


Ante-hoc Methods for Neural Networks

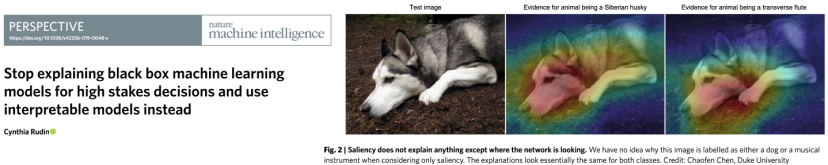
Learning goals

- Interpretability by sparsity
- Regularisation for interpretability
- Sequential feature selection

MOTIVATION



- Post-hoc methods do not always give you the entire picture
- Post-hoc methods are not always accurate
  - An explanation that is 10% inaccurate leads to lack of trust in the ML model
  - Hard to measure the accuracy of post-hoc methods
- Wherever possible use models that are interpretable-by-design



PERSPECTIVE
https://doi.org/10.1038/s42256-019-0048-x
nature machine intelligence

Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead

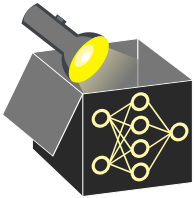
Cynthia Rudin

Test image | Evidence for animal being a Siberian husky | Evidence for animal being a transverse flute

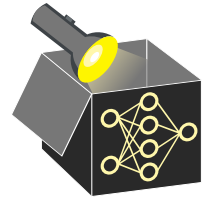
Fig. 2 | Saliency does not explain anything except where the network is looking. We have no idea why this image is labelled as either a dog or a musical instrument when considering only saliency. The explanations look essentially the same for both classes. Credit: Chaofen Chen, Duke University

SIMPLER MODELS

- Models that have an understandable decision-making process
- Models that have a smaller set of parameters or weights
  - Examples: Linear models, GAMs
- Models that have human-understandable decision structure
  - Examples: decision trees, random forests
- Models that have sparsity or only a few set of parameters or features that matter
  - Example: 1% of a large feature space, 1-hot encodings in language tasks



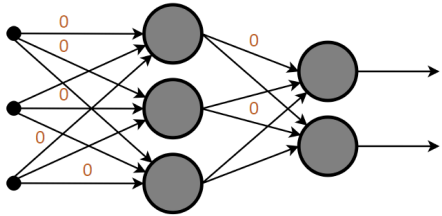
INTERPRETABLE BY DESIGN MODELS - SPARSE MODELS



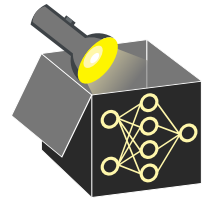
- Models that have explicitly enforce sparsity
  - through regularisation
  - through feature selection
- Sparsity through regularisation
  - E.g. L0, L1 regularisation
- Sparsity through feature selection
  - select a subset of impacting features for the prediction task

REGULARISATION IN NEURAL NETWORKS

- L0 norm is the number of non-zero parameters — setting weights to 0
- L1 sparsity — sum of the weights should be small



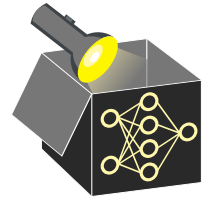
L1 REGULARISATION



- Optimising using L0 regularisation is hard
- L1 regularisation in neural networks can be achieved by gradient-based optimisation
- Degree of regularisation is a user-controllable parameter

$$\hat{\mathcal{L}}(W) = \alpha \|W\|_1 + \mathcal{L}(W)$$
$$\nabla_W \hat{\mathcal{L}}(W) = \alpha \mathit{sign}(W) + \nabla_W \mathcal{L}(W)$$

FEATURE SELECTION

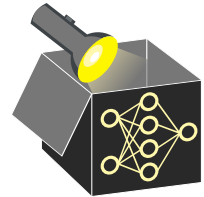


“Select a smaller features space which can efficiently describe the input data while reducing effects from noise or irrelevant variables and still provide good prediction results”

- Wrapper methods - Treat the model as a blackbox
- Filter methods
- Embedded methods
- Other methods

- Smaller feature space: subset of features, an embedded hyperspace

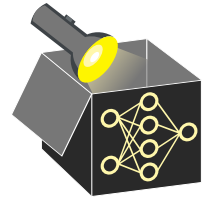
SEQUENTIAL FEATURE SELECTION



- Number of feature subsets is 2^N
- How do we reduce the computational complexity of checking each subset ?
  - Sequentially choose the most promising feature at each iteration

- Selection Set S= {}, All features N= $\{f_1, f_2, \ldots, F_n\}$
- In each iteration
  - compute utility of f- train a model with S $\cup\{f\}$ and measure validation perf.
  - terminate loop if no improvement of utility and return S
  - choose f in N/S that has max utility and add f to S

FEATURE SELECTION



- What are the short comings of sequential feature selection ?
  - Greedy might not be optimal
  - Global feature selection method
- How do we improve the greedy solution ?
  - Allow for backtracking, branch-and-bound
  - Use genetic algorithms GA, swarm optimisation
- How do we choose a local feature selection method ?
  - instance-wise feature selection methods